Transfer: momentum and impulse

antocciphysics | TEACHER KEY | CPA

Learning target: Apply momentum reasoning to new collisions and deflection cases.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 11; 3 course days later.*
Shift becomes 3 times as large: shift=Δv×time. The toy model omits orbital curvature/gravity and uncertainties.

02. Explain a worked example (worked)
Arrow goes from +6 to 0 kg·m/s; rectangle area -6 N·s.

03. Look and explain (model)
Total momentum 4 kg·m/s; after both move 2 m/s right. Kinetic energy is not conserved.

04. Build the solution (practice)
12 kg·m/s; 4 N opposite initial motion.

05. Use the idea (reason)
Compare total kinetic energy; must also match for elastic.

06. Choose a method (mixed) *Review: Unit 3, Day 2; 12 course days later.*
24 kg·m/s left.

07. Exit check (exit)
20 m; constant change/straight line approximation ignores orbital physics.

08. STRETCH: Apply it another way (stretch)
Examples 0.25 s at -24 N and 1 s at -6 N; same -6 N·s.

Teaching note: Self-contained daily transfer worksheet. Keep a separately assigned formal assessment independent.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 11; 3 course days later.*
$\Delta v \approx 0.0006$ m/s; impulse$\approx 4.2\times10^{5}$ N·s. Uses constant added transverse velocity over a straight-line comparison.

02. Explain a worked example (worked)
Arrow goes from +6 to 0 kg·m/s; rectangle area -6 N·s.

03. Represent and interpret (model)
Total momentum 4 kg·m/s; after both move 2 m/s right. Kinetic energy is not conserved.

04. Solve and justify (practice)
vf 3 m/s; initial KE 40.5 J; final 13.5 J; 27 J transformed.

05. Analyze the reasoning (reason)
Compare total kinetic energy; must also match for elastic.

06. Choose a method (mixed) *Review: Unit 3, Day 2; 12 course days later.*
v=-6 m/s; after reversal p=+24 kg·m/s.

07. Exit check (exit)
Impulse 6 N·s positive; initial p-2, final p 4; vf +2 m/s.

08. STRETCH: Apply it another way (stretch)
Impulse -3.2 N·s; average -40 N; triangle peak magnitude 80 N.

Teaching note: Self-contained daily transfer worksheet. Keep a separately assigned formal assessment independent.